



OPEN Advanced skin cancer prediction with medical image data using MobileNetV2 deep learning and optimized techniques

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Skin cancer, especially melanoma, has become one of the most widespread and deadly diseases today. The chances of successful treatment are greatly reduced if the melanoma is not treated in its early stages because it could spread aggressively. Hence, the diagnosis of skin cancer is very challenging as skin lesions are highly subjective to analyze and that type of expertise is exceedingly specialized. While there is an increase in the prevalence of skin cancer across the globe, there is an increase need of automated diagnostic systems that could aid medical personnel in making appropriate decisions within the requisite timelines. This study proposes construction of a deep learning model built on the MobileNetV2 architecture that has been memetic optimized for hyperparameter tuning. The memetic algorithm employs both global and localized search techniques to fine-tune the model parameters that include learning rate, batch size, and number of epochs to boost the efficacy of the model. This makes it possible for the proposed model to achieve high performance while remaining economical on resources. This makes the model suitable for real world clinical settings. The model achieved exceptional results, with 98.48% accuracy, 97.67% precision, and 100% recall, highlighting its strong ability to detect malignant lesions. The ROC AUC score of 99.79% further demonstrates its outstanding capability to differentiate between benign and malignant lesions. Notably, visualizations such as the Grad-CAM heatmap and Superimposed Image were crucial in providing interpretability to the model's decision-making process. The Grad-CAM heatmap highlighted the regions of interest in the lesions, showing how the model focused on key structural features. The Superimposed Image combined these heatmaps with the original lesion images, making it clear which parts of the lesions influenced the model's classification. These results underscore the potential of deep learning models, optimized with the memetic algorithm, to significantly improve skin cancer detection. By offering both high accuracy and interpretability, this model presents a valuable tool for dermatologists, facilitating faster and more reliable early diagnosis and ultimately improving patient outcomes.

Keywords Skin cancer, Deep learning, Memetic algorithm, MobileNetv2, Image prediction

Skin cancer is one of the most common and deadly diseases in the world, especially melanoma. It originates from skin cells' uncontrolled multiplication, primarily caused by ultraviolet (UV) radiation. If not detected at the right stage, it could cause metastasis¹. The higher the stage of cancer at time of diagnosis, the lower the chances of survival. Consequently, it is crucial that skin cancer be detected early². While there have been considerable developments in the field of dermatology, skin lesion analysis continues to be a significant barrier owing to its subjectivity and the level of skill involved³.

Machine learning and deep learning have transformed nearly every area of medical image analysis by improving the accuracy of diagnoses and prognosis⁴. Detecting malignant formations in medical images is an example. With the advent of digital pathology and radiology, the ability to accurately detect both benign and malignant lesions is crucial for early diagnosis, as well as devising an appropriate treatment modality⁵. In the last few years, deep learning methods, especially Deep Convolutional Neural Networks (CNN), have excelled in many areas of image classification, including medical imaging⁶.

With the use of a MobileNetV2 architecture memetic algorithm for hyperparameter optimization, this study presents a new paradigm for the classification of skin tumors focusing on the differentiation between the benign

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and malignant types. This approach optimizes the model accuracy and guarantees the model's morphology to the specifics of skin cancer images in a more effective and effective techniques during the differentiation of benign and malignant lesions in medical image analysis.

Materials and methods

To categorize the images of skin cancer as benign or malignant, a deep learning model based on MobileNetV2 architecture was used. The MobileNetV2 classification model is known to achieve a high level of accuracy while also being cost-effective, thus it was used in this framework to guarantee accurate classification with very little overhead⁷. Furthermore, a memetic algorithm was also incorporated to fine-tune the model's hyperparameters. This type of optimization is more sophisticated as it employs a combination of global and localized methods based on evolution to improve the model in question^{8–11}. This particular framework was constructed using the readily available deep learning framework, PyTorch. The methods including dataset preparation, model architecture, training, and optimization plans will be explained in the following sections.

Dataset construction and preprocessing

The dataset in this study has images of patients with benign or malignant conditions. Each image is stored as a .jpg file, and the model aims to classify the images as either benign or malignant.

Initially, the dataset contained 30 benign and 240 malignant samples, resulting in an imbalanced distribution. Due to the imbalance between benign and malignant samples in the dataset, data augmentation techniques were applied to increase the number of benign images. This process was done through the ImageDataGenerator class which performs transformations of the original images¹². These transformations include rescaling, rotation, shearing, zooming, shifting in height and width, and flipping horizontally. To address the class imbalance, data augmentation was applied to benign images using the following parameters: `rescale = 1./255`, `rotation_range = 20`, `width_shift_range = 0.2`, `height_shift_range = 0.2`, `shear_range = 0.2`, `zoom_range = 0.2`, `horizontal_flip = True`, `fill_mode = 'nearest'`. These transformations increased the variability of benign samples and helped improve the model's generalization. The goal was to produce enough images so that the dataset would be balanced. These created images were stored in a specific folder and were used in the other analyses that followed. As a result of this process, the number of benign images was increased to 200.

Utilizing PyTorch's, a custom Dataset class was created to manage the dataset. This class makes use of two folders where images are stored, takes images from both folders, assigns the appropriate label (0 for benign and 1 for malignant), and applies the required transformations. The transformations involve changing the size of the images to a dimension of 224×224 pixels, changing the images into tensors, and normalizing the images with ImageNet's standard mean and standard deviation⁶. This normalization guarantees that the inputs to the model are as close to 0 as possible and have a standard deviation of 1, which allows the model to train more easily.

Using a random split, the dataset is divided into three sets: training (70%), validation (15%), and testing (15%) samples. Each folder is ensured to contain a representative mixture of both benign and malignant samples.

Model Architecture- MobileNetV2

The architecture was built on MobileNetV2, a lightweight CNN tailored for mobile and edge devices. The model is based on MobileNetV2, which is a lightweight CNN architecture made for mobile and edge devices with limited processing capabilities. MobileNetV2 improves upon the previous MobileNet V1 architecture and enhances accuracy as well as efficiency of the model. MobileNetV2 is designed in a way that allows it to use depth-wise separable convolutions, which allow spatial filtering to be decoupled from this feature processing. This decoupling helps MobileNetV2 achieve drastic reductions in model parameters and computational complexity making it suitable for deployment in resource constrained environments without compromising model performance. In those separable convolutions, each input channel is first filtered separately by a single filter, then a pointwise 1×1 convolution is used to combine the outputs. This computation reduces the number of processes needed greatly, and as a result, make MobileNetV2 the best appropriate for tasks where processing power is limited, like model pruning for medical image analysis⁷.

In MobileNetV2, the most innovative feature is the addition of inverted residual blocks. Unlike traditional residual blocks that features two convolutive layers for dimensionality reduction and increase, inverted residual blocks add a pointwise convolution for feature dimensionality expansion, followed by a depth-wise separable convolution for spatial relationships, and lastly another pointwise convolution for dimensionality reduction. Such unique architecture enables the model to extract complex features without overly increasing computational cost. This makes MobileNetV2 highly efficient for detailed classification tasks like skin cancer detection which requires high accuracy while using minimal resources¹³.

Additionally, the architecture incorporates linear bottlenecks in each inverted residual block. Linear bottleneck layers eliminate the issue of losing critical features during dimensionality reduction. The use of linear activations in the bottleneck layers serves to reduce non-linearity, and therefore avoiding excessive computation. This design allows for the improvement of network speed and efficiency while increasing responsiveness, particularly important for real-time diagnostic tools where memory and processing power is limited.

The model's initial training was performed using the ImageNet dataset, which served as a strong feature extractor for the binary classification task in skin cancer detection¹⁴. By applying transfer learning from ImageNet, the network was able to utilize general visual features that were available and could be adjusted over the specific medical dataset, which improved the generalization of the model and the speed at which it converged. This pretraining helped in improving the model's ability to extract image skin lesion classification features, which reduced the need to train the model from the scratch in the first place⁷.

Model training and memetic algorithm for hyperparameter optimization

This study used a memetic algorithm in tuning the model skin cancer classification terms in order to boost the reliability of the model and performance. Memetic algorithms are a type of population-based search technique where a genetic algorithm is combined with local search to bring about a new level of evolution known as cultural evolution⁸. This enables a more efficient exploration of a given search space, because the solutions are first globally optimized (using genetic operations), and then locally refined (using local search procedures).

Mechanism of the memetic algorithm

As with any other algorithm, memetic procedures could be divided into various steps for efficient functional completion. It generally has two phases, which include: (i) global search, and (ii) local refinement.

Global Search Phase: A population of candidate solutions (superparameter combinations) is generated. Each candidate constitutes a potential amalgamation of phantom parameters, for example: optimal learning rate, batch size, number of epochs summations, and type of optimizer. These candidates are subjected to genetic operations like selection, crossover and mutation^{8,9}.

- o Each candidate is evaluated through selection, putting them into a fitness test which is determined by how well they performed in the validation set. Each selected candidate goes through reproduction, where the best candidates are selected through a 'best performing candidates' policy.

- o Crossover merges parts from two or more parent solutions to produce offspring that have the combined features of both parents.

- o A mutation makes slight random changes to a potential solution so that genetic diversity is increased, enabling the algorithm to explore new areas of the solution space.

The global search phase is aimed at efficiently covering a broad space of possible solutions by creating many diverse candidates that can yield high performing hyperparameters.

Local refinement phase: After the candidate solutions have been created through biological processes, each candidate is taken through a local search. This candidate solution is improved or optimized by using a more precise technique. In most cases, iterative optimization techniques such as gradient descent or hill climbing are used. The essence of the local search is to build upon the existing solution and improve the candidate's performance. This process further optimizes the high performance score solutions to add more granularity in finding local optimums^{8–10}.

The combination of global search and local refinement in the memetic algorithm gives it advantages over standard optimization methods. Genetic algorithms operatively on a broad solution space, optimize in every direction below mistakes and guarantee that as soon as great solutions are identified, the results are further enhanced.

Application in hyperparameter optimization

Within the parameter of a skin cancer classification model, a memetic algorithm was used in order to optimize hyperparameters that are essential to the model's accuracy and efficiency^{8–11}. Such hyperparameters were optimized by using a memetic algorithm:

- **Learning Rate:** This is a hyperparameter that dictates the size of the steps taken in an iteration towards a minimum. When the learning rate is too high, the model overshoots the optimal values, while a low learning rate results in slow convergence.
- **Batch size:** This parameter sets the amount of training examples that will be utilized to calculate the gradient once as the model parameters are updated. An increase in the batch size may result in more stable training, however, a decrease in the batch size leads to more variability, but faster convergence.
- **Number of epochs:** This is the amount of scheduled passes through the entire training dataset. There may be an excessive number of epochs, which could lead to overfitting, or insufficient ones resulting in underfitting.
- **Optimizer Type:** The selection of an optimizer plays a considerable role in model performance for instance, Adam and SGD (Stochastic Gradient Descent).

Adam is an enhanced form of SGD since it adapts the learning rate for every distinct variable based on the first moment and second moment of the gradient. The said techniques offer increased stability and greater speed of convergence without the difficulty of setting a learning rate. Adam has become a standard choice in training deep neural networks¹⁵. However, in comparison to Adam, SGD is used as a baseline optimizer to check the performance against Adam¹⁶.

The memetic algorithm led to the discovery of an efficient learning hyperparameter in the following order:

- Learning rate: 0.000971.
- Batch size: 64.
- Number of epochs: 10.
- Optimizer type: Adam.

During optimization, it was noted that the learning rate was most sensitive to constraint in the gradient descent changes, especially using the Adam optimizer where the rate was able to learn faster for each variable. The batch size of 64 was selected to provide optimal computation with minimal risk of overfitting the model. The number of epochs was lowered as well, down to ten in order to combat overfitting, allowing to better generalize the model in the end.

Benefits of memetic optimization in hyperparameter tuning

The use of memetic algorithms for hyperparameter optimization allows for an easier approach when compared to grid search or random search. The large-scale set of populations allows for better coverage of the hyperparameter space. Unlike grid search which verifies every hyperparameter-set combination or random search that does not focus on critical areas, the memetic algorithm improves the search direction through global probing and local focusing.

Moreover, the advanced stages of the optimization routine improved the model by increasing some selected hyperparameters. This optimization-outline ensures that hyperparameters of the model are not just selected based on performance but tailored for maximization where optimal performance and good general structural behavior are obtained^{8–11}.

Training and evaluation process

The training process is considered to be the same as any other model in which training occurs over several epochs with mini-batches of data. During each epoch, the loss is calculated from the model's predictions, and the optimizer uses the estimated loss to update the model's parameters. Model accuracy is calculated after each epoch using the accuracy on the validation set. Training continues for the specified number of epochs or until convergence is achieved.

After the model is trained, its evaluation on the test dataset is performed to analyze its performance on generalization ability. In addition, ROC curves are plotted to visualize the relationship between true positive rate and false positive rate at different decision thresholds¹⁷.

Visualization of results

Numerous visualizations are included in order to provide more context to the model performance:

- The Confusion Matrix: This heatmap represents the count of true positive, false positive, true negative, and false negative results, as it relates to the errors analysis the model makes¹⁸.
- Top 10 Predictions: A 100% certain predictions is visualized by the top ten most confident correct predictions to evaluate on the factors that influence the model decisions.
- ROC Curve: Combined ROC curve for both the training and validation sets is plotted with the area under the curve (AUC) as a marker of the model's ability to discriminate¹⁷.
- Grad-CAM Visualization: To Integrate Faster Vision Transformers to medical image classification models, a Grad-CAM class activation mapping technique was used to analyze the model decision making process. Grad-CAM is a visualization technique that indicates which parts of an image input were most important in changing the model outcome. It achieves this by calculating the gradients of the output with respect to the activations of the output of the last convolutional layer. The resulting gradients are multiplied by the activations and summed up to form a weighted average. This image is then used to create a heatmap which indicates the most decisive features in the model input. The heatmap is superimposed on the original image to the most important regions of the image for the model decision are emphasized¹⁹.

All image processing and visualizations were performed using Python (version 3.9) in Visual Studio Code.

Evaluation metrics

The success of the model is determined through the following evaluation metrics:

Accuracy: the ratio of correct predictions to total predictions made²⁰.

Precision: the ratio of relevant predictions (true positives) to the total of positive predictions made²⁰.

Recall: the ratio of relevant predictions (true positives) to the total number of actual positives²⁰.

AUC: the ROC curve, it describes the ability of the model to separate benign and malignant cases¹⁷.

The model is implemented in PyTorch, while the training is executed on a machine equipped with a GPU for faster processing when it is available. This infrastructure provides a strong and automated solution for skin cancer detection by applying deep learning methods.

Results

The evaluation of a proposed skin cancer classification model which uses MobileNetV2 in combination with a memetic algorithm for hyperparameter optimization is done through different key metrics. It is noted that the model has an accuracy of 98.48%, meaning that almost 98.5% of the instances in the dataset were correctly classified. Precision was determined to be at 97.67%, which illustrates that when the model has predicted malignant lesions, it is correct roughly 98% of the time which is vital in minimizing false positives and unnecessary procedures. The models recall score of 100% signifies that the model is able to capture all actual malignant lesions, further proving the model's sensitivity towards critical cases is excellent. The balance between precision and recall is further illustrated by the models F1 score of 0.9882, which guarantees accurate detection and minimal false positives or negatives. Lastly, confirming the model's ability to differentiate lesions from each other, the ROC AUC score is at 99.79%. These outcomes illustrate the model's accuracy and reliability which make clinical dermatology applications, particularly in diagnosing skin cancer, to be also rapid and efficient.

In Fig. 1 we highlight the classification results of the model's prediction on both malignant and benign skin lesions. Notably, the model's predictions were found to perfectly match the ground labels. The correctly classified malignant lesions feature other morphologically distinct patterns suggesting the model's generalizability in different lesion types. Similarly, benign lesions of various textures and color distributions were correctly recognized, demonstrating the effectiveness of the classification framework.

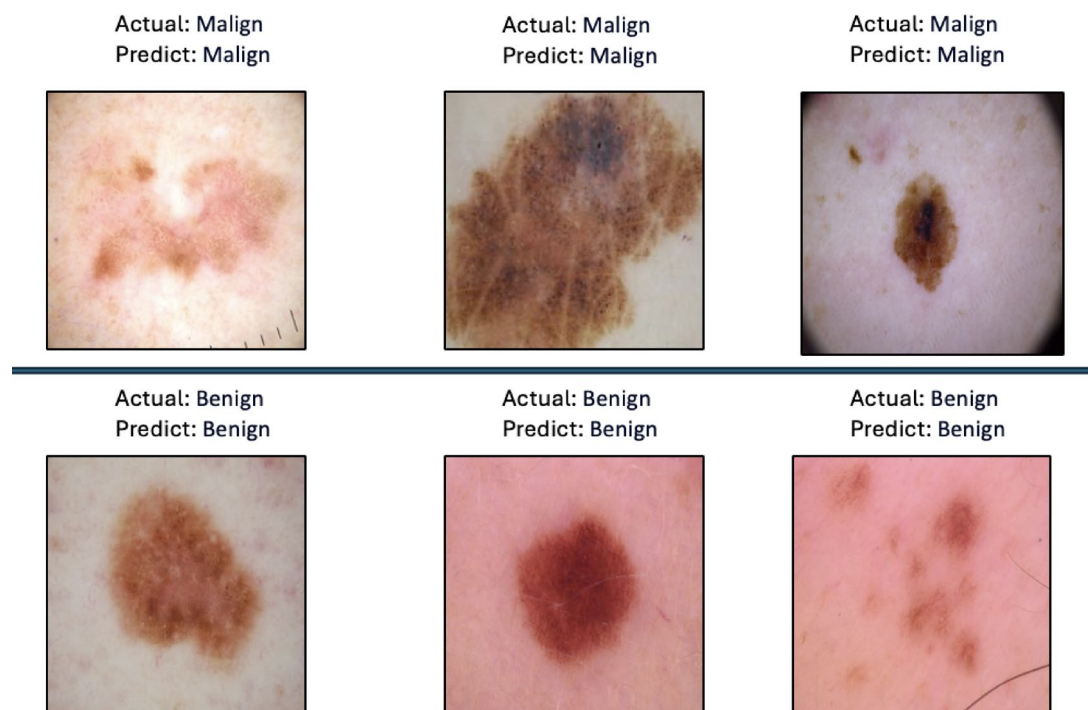


Fig. 1. The data written as Actual are images taken from the Kaggle dataset.

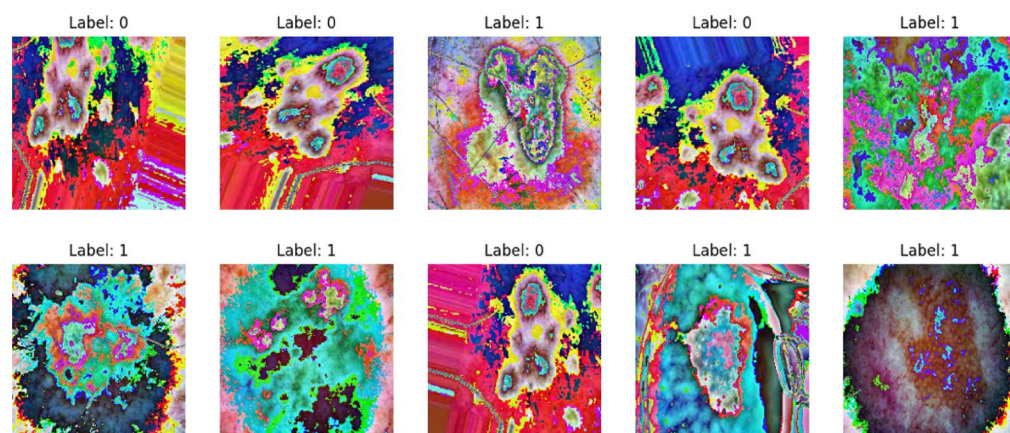


Fig. 2. Grad-CAM visualizations; Label 0: Benign Lesions, Label 1: Malignant Lesions.

Figure 2 showcases the Grad-CAM visualizations, which provide an interpretable assessment of a deep learning model's classification accuracy of skin lesions into two categories: benign (0) or malignant (1). The highlights reveal areas that the model regarded as most important in forming its predictions.

For the benign lesions (0), the model mostly zoomed in the center regions of the lesions while largely ignoring the surrounding tissues. This points to the fact that the classification was mostly based on the core structural features of the lesion. However, for malignant lesions (1), the activation maps indicate broader and more intense focus areas than those evident in the center of the lesion. This suggests that the model was able to detect more complex structural patterns, which are typically associated with cancerous lesions.

Figure 3 depicts the heatmap and its corresponding overlay image that provides an interpretable explanation to the model's prediction process for skin lesions classification. The model exhibited regions with higher activation in the heatmap, which indicates the area of primary importance for its decision-making process.

In the superimposed image, the heatmap is applied to the original skin lesion figure so that it can be seen how the model was able to identify and localize regions of interest for the lesion's classification. The primary concentration of high-activation areas, marked in red, which typically are critical features that differentiate benign from malignant lesions, is dominated around the borders and central structures of the lesions.

Figure 4, the ROC curve analysis indicates that the model had exceptional performance distinguishing between the two sets of skin cancer cases. On the training dataset, the model with an AUC of 99.12% showed

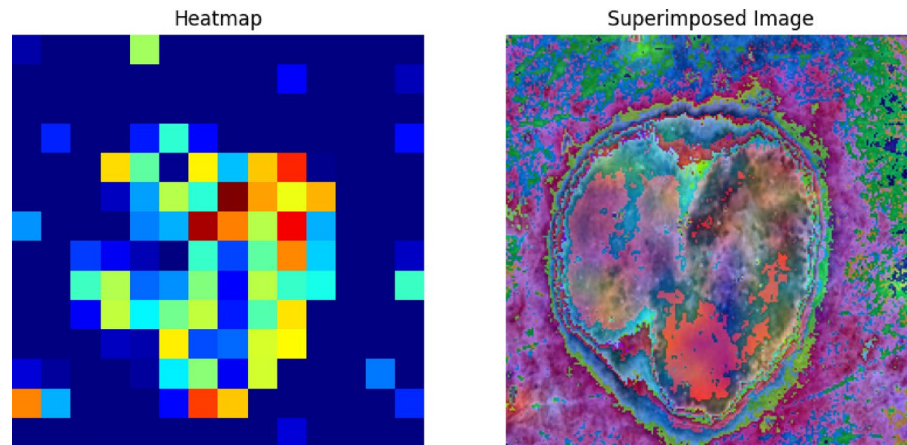


Fig. 3. Heat Map and Superimposed Image.

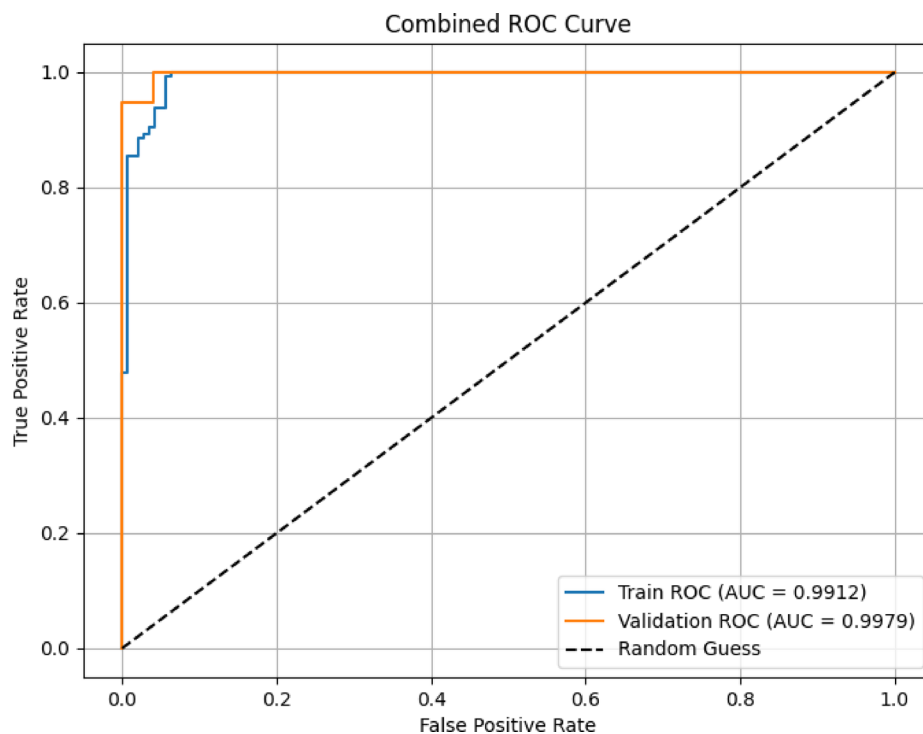


Fig. 4. ROC-Curve Graph.

remarkable discriminatory capability. On the other hand the validation dataset achieved even greater AUC of 99.79%. These results indicate for the excellent generalization performance of the model. These results strongly support the clinical applicability of the model in dermatological diagnostics because it gives reliable and precise predictions for real-world use.

Figure 5, the proposed model's ability to discriminate between benign and malignant lesions is detailed in the confusion matrix. Out of the total 26 identified benign cases, 25 were correctly classified, with only a single case of false positivity. Furthermore, all 38 malignant cases were identified, achieving a 100% recall for malignant lesions. These results show that the model is accurate in classifying negative cases, which is essential in clinical settings; missing malignant cases could be hugely detrimental to patients. The model's low false positive rate also ensures that unwarranted invasive procedures are kept to a minimum. The accuracy of the model speaks to its effectiveness and reliability in diagnosing skin cancer.

Figure 6 shows the change in accuracy through the cycles of training. During the initial epochs, there is a slight decrease of accuracy during training. However, the model soon recovers from the decline and performs similarly to the provided validation accuracy. The close alignment of training and validation accuracies suggests that the model does not overfit the data and generalizes effectively to new data.

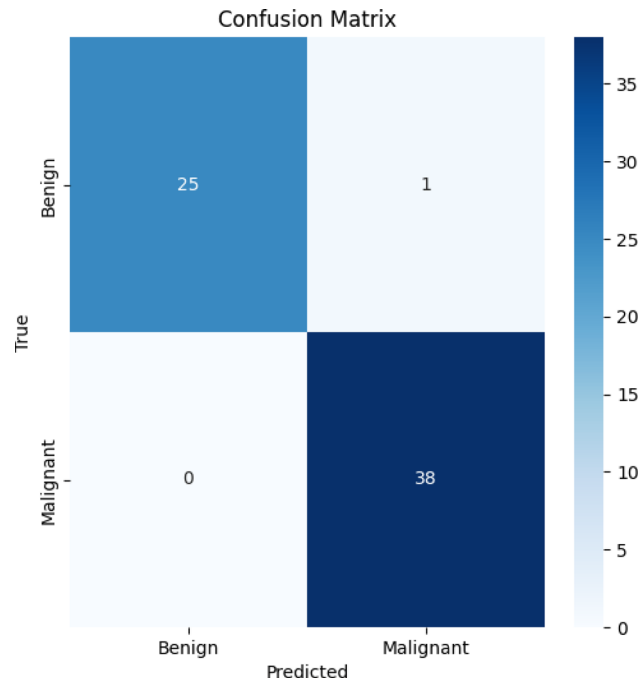


Fig. 5. Confusion Matrix.

Epoch	Training Accuracy (mean ± sd)	95% CI (Training)	Validation Accuracy (mean ± sd)	95% CI (Validation)
1	0.80 ± 0.02	0.7608–0.8392	0.79 ± 0.01	0.7704–0.8096
2	0.68 ± 0.03	0.6212–0.7388	0.97 ± 0.01	0.9504–0.9896
3	0.78 ± 0.01	0.7604–0.7996	0.97 ± 0.008	0.9543–0.9857
4	0.94 ± 0.02	0.9008–0.9792	0.97 ± 0.005	0.9602–0.9798
5	0.96 ± 0.01	0.9404–0.9796	0.97 ± 0.004	0.9622–0.9778
6	0.95 ± 0.015	0.9206–0.9794	0.97 ± 0.003	0.9641–0.9759
7	0.96 ± 0.01	0.9404–0.9796	0.965 ± 0.003	0.9571–0.9729
8	0.95 ± 0.02	0.9108–0.9892	0.965 ± 0.003	0.9571–0.9729
9	0.96 ± 0.01	0.9404–0.9796	0.965 ± 0.002	0.9611–0.9689
10	0.965 ± 0.008	0.9493–0.9807	0.965 ± 0.002	0.9611–0.9689

Table 1. Accuracy metrics per epoch (Mean ± SD, 95% CI).

The training and validation accuracy values along with their standard deviations and confidence intervals for each epoch are summarized in Table 1.

Figure 7 depicts a graph that outlines the training loss. The graph shows a high slope in the early epochs followed by a plateau. This significant decline supports how effective the Memetic algorithm excels in the optimization landscape and how accurately it adapts the training data.

Figure 8, the validation loss shows a consistent declining behavior which contains a converging value parallel to the loss in training. The unified decrease in training and validation losses proves that the model is able to generalize successfully without overfitting.

Discussion

The high prevalence of skin cancer and its ability to be life threatening makes early diagnosis essential which further signifies the need for accurate tools¹. Innovations in image deep learning alongside optimization in the last few years has changed the landscape of medical image classification, particularly skin lesion imaging, resulting in more accurate results. In relation to this, the study aims to resolve the issue of differentiating between malignant and benign skin tumors using advanced classification algorithms. The methodology seeks to achieve greater diagnostic accuracy using a high performing convolutional neural network while also optimizing for adaptable to the complex nature of medical imaging datasets.

It is probable that the use of memetic algorithms integrating hyperparameter optimization contributed to strong results by optimizing the learning rate, batch size, and epochs. Memetic algorithms that encompass global search and local search are well recognized for their efficiency in utilizing large parameter scope, leading to improved model performance⁸. This approach allowed skin cancer images' hyperparameters to be adjusted



Fig. 6. Training and Validation Accuracy.

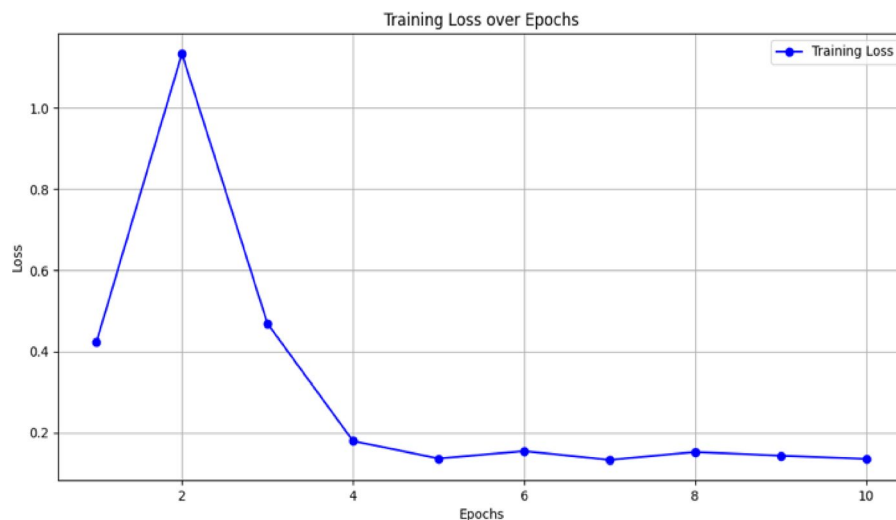


Fig. 7. Training Loss Over Epochs.

for the maximum generalization and sensitivity to the images' details. At the same time, it ensured that the model obtained impressive results without the burden of excessive computation, making it sensible to be used in clinical practice⁷.

During the initial experiments, the model was trained for 50 epochs. However, after the 10th epoch, the model began to overfit, resulting in a consistent output of 1.00 for predictions. This behavior indicated that the model was overly specialized to the training data, failing to generalize to new, unseen examples. Overfitting is a common issue in deep learning, particularly when training over extended epochs with a relatively small dataset or without proper regularization techniques. To mitigate this, we reduced the number of epochs to 10, which significantly improved the model's ability to generalize while maintaining high accuracy. This adjustment highlights the importance of balancing training duration and model complexity to prevent overfitting and ensure optimal performance in real-world applications.

The visual data from Grad-CAM suggests that the model was able to learn distinct focus strategies for the two classes, which is critical for classifying cancers accurately for both benign and malignant lesions. These findings support the model's ability to detect important clinically relevant and significant regions of interest and further strengthen the trust in the model's reliable predictions like other previous studies using Grad-CAM for model interpretability^{21–25}.

Superimposed shows that the model makes correct classifications and pays attention to the regions that are medically relevant. The combination of the superimposed and heatmap image confirms that the deep learning framework is able to incorporate both the spatial and the structural context of the image, which increases its diagnostic value. These outcomes offer evidence to support the claim of the model's interpretability and its usefulness in practical clinical settings as previously noted with the use of superimposed images to increase the interpretability of the model^{26–29}.

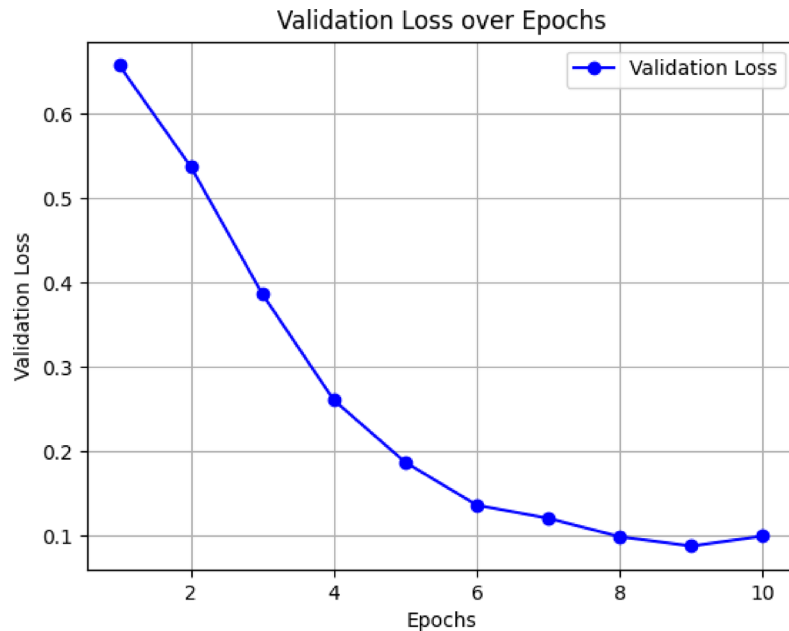


Fig. 8. Validation Loss Over Epochs.

Mahbod et al. performed the task on an ensemble of pre-trained CNNs with the skin lesion classification of melanoma, seborrheic keratosis, and benign nevi, where they were managed to get AUC scores of 84.8% and 93.6% on binary classification problem of melanoma and seborrheic keratosis, respectively³⁰. In our case, the model was built around the idea of binary classification, which resulted in much better results with the ROC AUC score of 99.8% and the accuracy of 98.48%. Their method was successful in multi-class classification, but the combination of MobileNetV2 with memetic optimization in our case outperformed them in a higher sensitivity and specificity in classifying benign versus malignant lesions. These results suggest the need to improve the method of computer-aided diagnosis of skin diseases by the employing more sophisticated optimizing algorithms to improve accuracy for specific classification tasks.

Priyeshkumar et al. proposed a new model called Multi-Grained Enhanced Deep Cascaded Forest (Mg-EDCF) for skin lesion classification into seven different classes based on the HAM10000 dataset which achieved 98.19% accuracy³¹. Their approach of multi-grained scanning and Random Forest feature generation was able to surpass established models like ResNet, AlexNet, and VGG16 outperformed more established models such as ResNet, AlexNet, VGG16, etc. While their method deals with a complex multi-class problem, our binary classification model offers comparable results but with even higher accuracy at 98.48% which shows that it is more effective in detecting skin cancer.

Sarwar et al. developed a new model with ResUNet architecture combined with Ant Colony Optimization for skin lesion binary classification achieving accuracy of 95.8%, Dice coefficient of 93.1%, and Jaccard index of 87.5%. Their approach effectively uses deep learning with other optimization techniques to increase diagnostic accuracy³². Compared to that, our implementation using MobileNetV2 and memetic optimization outperformed both in accuracy and ROC AUC. Although both studies place significant focus on the importance of optimization of deep learning models, my approach provides slightly better overall performance and therefore greater potential for clinical implementation.

Gayatri and Aarthy's study resolved the challenge of class imbalance and overfitting using a fine tuned ResNet-50 model, along with data augmentation and focal loss. It achieved an impressive 98.85% accuracy, 95.52% precision, and 95.93% recall³³. In contrast, our approach employs a MobileNetV2 model optimized via a memetic algorithm for hyperparameter tuning. This model performs competitively while maintaining high sensitivity and specificity, which indicates that it can be very effective for skin cancer diagnosis. This confirms the finding of the preliminary research which underscored the pertinence of applicable deep learning techniques in addressing the data imbalance phenomenon. By effectively condensing the results of both, it demonstrates how the metrics for construction classification models in clinical dermatology gets simplified.

Mateen et al. propose a hybrid deep learning framework that integrates U-Net, Inception-ResNet-v2 and Vision Transformer with a self attention module for diagnosis of melanoma. Evaluated on the ISIC2020 dataset, it achieved 98.65% accuracy, 99.20% sensitivity, 98.03% specificity which surpassed the benchmark approaches [34]. Our model, on the other hand, opts for MobileNetV2 optimized through a memetic algorithm. In all, this tends to aid in the increase of performance, but at the same time, the primary focus is on the reduction of resource consumption. Both studies make a case for the importance of hybrid deep learning and its optimization on the detection of melanoma even more so for fast tracking clinical application like ours.

Ogundokun et al. designed a deep learning model that uses a combination of Xception and MobileNetV2 to detect skin cancer. It achieved an accuracy of 97.56%, precision of 93.33%, and recall of 100%. On the other hand, their MobileNetV2 model had an accuracy of 90.24% and recall of 82.35%, showing what a hybrid model can do

in terms of performance improvement³⁵. In the same way using MobileNetV2, we were able to classify malignant and benign lesions with greater accuracy and recall than what their study reported MobileNetV2 achieved, which was exemplary. These findings further corroborate that the choice of model optimization techniques and model combinations is critical in improving skin cancer detection systems. Both studies highlight the importance of optimizing and selecting a suitable model to improve classification and diagnosis accuracy.

Lee et al. introduces the AGS network that adds Augmentation (A), GAN (G), and Segmentation (S) modules to automate skin cancer classification, especially in the presence of small datasets. Their methodology has been tested with eight different deep learning classifiers, and the results are dramatically improved. The ResNet50 + AGS has an accuracy of 95.87% and F1-score of 95.73%. This method is the ideal demonstration of achieving an increased model performance through the combination of numerous augmentation methods³⁶. On the other hand, our model seeks to accomplish MobileNetV2 optimization through memetic algorithm hyperparameter tuning in order to automate the classification of malignant from benign lesions, achieving excellent results. Both studies contribute greatly towards the understanding of skin cancer detection optimization and augmentation strategies, while Lee et al.'s work concentrates on the application of more techniques for enhanced performance³⁶.

Alotaibi and AlSaeed (2025) integrate attention mechanisms with Xception transfer learning model and propose deep learning method with the intent to recognize skin cancer. Their research claims that the addition of Self-Attention, Soft-Attention, and Hard-Attention to the Xception model improves accuracy in the model performance and achieves the highest accuracy of self attention at 94.11%³⁷. This method is also remarkably effective in recall, which is very important in medical diagnosis. Conversely, our model, which is based on MobileNetV2 optimized with memetic algorithm for hyperparameter tuning, provides excellent results in the classification of malignant and benign lesions. Optimization techniques are important aspects in both studies, and while Alotaibi and AlSaeed did improve feature selection using attention mechanisms, which stood out in the former study, hyperparameter optimization techniques utilized in classification accuracy more than emphasized in the other study³⁷.

The model that was studied compares with one by Deuſing et al., which describes the diagnostic accuracy of confocal laser scanning microscopy (EVCN) of melanocytic lesions. The EVCN method obtained accuracy rates of 67.7%–79.2% based on the level of the investigator. The model performed significantly better than that. By utilizing hyperparameter tuning on MobileNetV2, the deep learning model was capable of reach an astonishing accuracy of 98.48%. The model with higher precision and recall was the one having a ROC AUC of 99.79%. Therefore, it could be concluded that the model has the ability to distinguish between benign and malignant lesions with great accuracy. As earlier mentioned, EVCN allows for detailed analysis, however results could differ greatly from one clinician to another due to reliance on individual interpretation. In comparison, the model make eases produces predictions and removes the variability that comes with diagnosis³⁸. Both methods discussed highly effective, however, the high accuracy and scalability of the model means that the clinical settings that need a quick and dependable automated diagnosis will find it most useful.

Unlike the approach of Hamim et al., where DenseNet121 is combined with XAI techniques for interpretability, the focus of our model is on obtaining high diagnostic accuracy. SmartSkin-XAI achieved an accuracy of 97% on the ISIC dataset and 98% on the Kaggle dataset while striving for interpretability and performance [39]. In contrast, our model utilizing MobileNetV2 with hyperparameter optimizations achieved an accuracy of 98.4%, very high precision and recall rates of 97.67% and 100% respectively, and an ROC AUC of 99.79%, surpassing SmartSkin-XAI in the performance of differentiation of benign and malignant lesions. While both models aim to improve decision making at the level of the clinician, SmartSkin-XAI is claimed to deliver better results in interpretability which in turn builds clinician's trust at the expense of our model which is more accurate and scalable. As discussed in the previous comparison, both models have their merits and demerits that impact the efficacy of using an AI based skin cancer diagnosis solution.

As previously stated, Kreouzi et al. (2024) analyzed CNN models designed to focus on melanoma detection and its uses, similar to the approach taken in our study. Certainly, it suggests that deep learning has potential usability for classifying skin cancers. However, their study put more emphasis on the tradeoff between accuracy and model efficiency, particularly for mobile, resource-limited operations. In contrast, Kreouzi et al. work put greater focus on accuracy and model efficiency during mobile and resource-limited functionality. Our model places utmost importance on diagnostic accuracy as well as reliability when it comes to clinical settings. Kreouzi et al. showed MobileNetV2 indices good inference time and model size so it is suitable for practice⁴⁰. Nevertheless, our model, which is built on a MobileNetV2 augmented with a memetic algorithm for hyperparameter tuning, achieves superior stratification of benign and malignant lesions while providing better overall diagnosis. In short, while both models serve the purpose of enhancing melanoma detection, achieving accuracy and precision within clinical dermatology remains the primary focus for which our approach is geared towards.

Both our study and Mendi et al. have a common objective in AI application for melanoma detection, yet they diverge in critical aspects of their foci. Mendi et al. discuss AI-assisted non-invasive imaging methods, including dermoscopy and confocal microscopy, which have the potential to elevate the accuracy and feasibility of diagnosis⁴¹. Alternatively, our work is focused on combining MobileNetV2 and memetic optimization, achieving exceptional results in skin cancer classification.

Artificial intelligence applied to the study and imaging of skin conditions not only augments diagnostic accuracy, but also possesses the tremendous opportunity to building novel solutions demonstrating the impact of AI in greatly these fields^{42–52}.

Conclusions

The study highlights the construction of a model utilizing deep learning technology for skin cancer detection and its proven accuracy and classification proficiency. The model's performance, which was significantly improved through optimization strategies such as the memetic algorithm, suggests that it can provide considerable support

to dermatologists for early clinical diagnosis. The importance of diagnosing patients accurately and at an early stage cannot be overstated, and with those factors in mind, it would greatly benefit patient outcomes. As for future studies, focusing on using other cancer types to strengthen the model with data augmentation could be helpful.

Limitations

This study presents a robust computational framework for skin lesion classification using a memetic-optimized MobileNetV2 model; however, it is subject to several key limitations. Notably, the small size of our dataset, coupled with its open-access Kaggle origin, inherently restricts detailed provenance, comprehensive metadata (such as imaging modality or Fitzpatrick skin type), and pathological confirmation, which are crucial for real-world clinical applicability. Furthermore, the limited test set size and reliance on a single train-validation-test split prevented the implementation of more robust validation techniques like k-fold cross-validation, potentially leading to inflated performance metrics. While data augmentation was applied to address class imbalance, the absence of other strategies like weighted loss functions is a recognized constraint. The lack of explicit early stopping mechanisms and clinical dermatologist validation of Grad-CAM visualizations also impacts the interpretability and direct clinical relevance of our findings. Finally, limited computational resources and the absence of dedicated dermatology expertise within the study team restricted our ability to validate the model on broader datasets and compare it against real-world clinical variability. Future work will focus on addressing these limitations to enhance the generalizability and clinical validity of our model in practical healthcare settings.

Data availability

The dataset is available at <https://www.kaggle.com/datasets/shashanks1202/skin-cancer-dataset>.

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Author contributions

T.Ö. contributed to the study conception and design. Material preparation, data collection, and analysis were performed by T.Ö. The first draft of the manuscript was written by T.Ö., and the review draft was written by T.Ö. and N.V.K. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval

Since it is an open access dataset, ethics committee approval was not required.

Consent to participate

Since it is an open access data set, there is no voluntary consent form.

Consent to publish

All images in the article were created by us using Python.

Additional information

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